WEEK 10 OF 15 · STAGE 2: GCAM IN DEPTH

Emissions, carbon removal and policy

Set a price or set a limit: two doors to the same room

New R skill gcamdata: finding a chunk, editing an assumption, running the driver

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 **Foundations**	Week 1 Why model the future?	Week 2 Climate primer	Week 3 Energy primer	Week 4 Economics primer	Week 5 Cost-benefit IAMs: DICE	Week 6 The scenario framework
Stage 2 **GCAM in depth**	Week 7 Inside GCAM	Week 8 Energy in GCAM	Week 9 Land, water and bioenergy	**Week 10 Emissions and policy**		
Stage 3 **Advanced methods**	Week 11 Optimisation IAMs	Week 12 Why models disagree	Week 13 Uncertainty	Week 14 Frontiers and limits	Week 15 Projects and review	

TODAY'S GOALS

By the end of today you can

1. Trace how GCAM turns activity into emissions and Hector turns emissions into warming
2. Explain the mirror between a carbon price and an emissions limit
3. Describe three ways GCAM removes CO2 and what each one costs the rest of the model
4. Run policy scenarios and change one assumption in gcamdata

? **Big question** Should a government say 'emissions cost $50 a tonne' or 'emissions must reach zero by 2050', and what does a model do with each?

Two ways to tell a model about climate policy

Option A: 'Every tonne of CO2 costs $50.' Option B: 'Emissions must be zero in 2050.'



With option A, what does the model work out? With option B?

HOW IT WORKS

From activity to temperature

1

Activity

Fuel burned, land cleared, goods made, animals raised

2

Emissions

Activity × emission factor, less any abatement

3

Hector

Turns emissions into concentrations and forcing

4

Temperature

Global mean warming for the scenario

Steps 1 and 2 happen inside GCAM; steps 3 and 4 are Hector, the model you ran in week 2

Worked example: counting emissions

Problem Emissions = activity × emission factor × (1 − abatement share). A region makes 1,000 TWh of coal power; a landfill emits 2 Mt of methane.

1. Coal power factor: 0.9 Mt CO2 per TWh (0.9 t per MWh)
2. CO2 = 1,000 × 0.9 = 900 Mt
3. A carbon price makes 30% methane capture worthwhile (cost curve)
4. Methane = 2 × (1 − 0.30) = 2 × 0.7 = 1.4 Mt

Answer

900 Mt CO2 from power; methane falls from 2 to 1.4 Mt

AN EVERYDAY PICTURE

A toll road or a gate

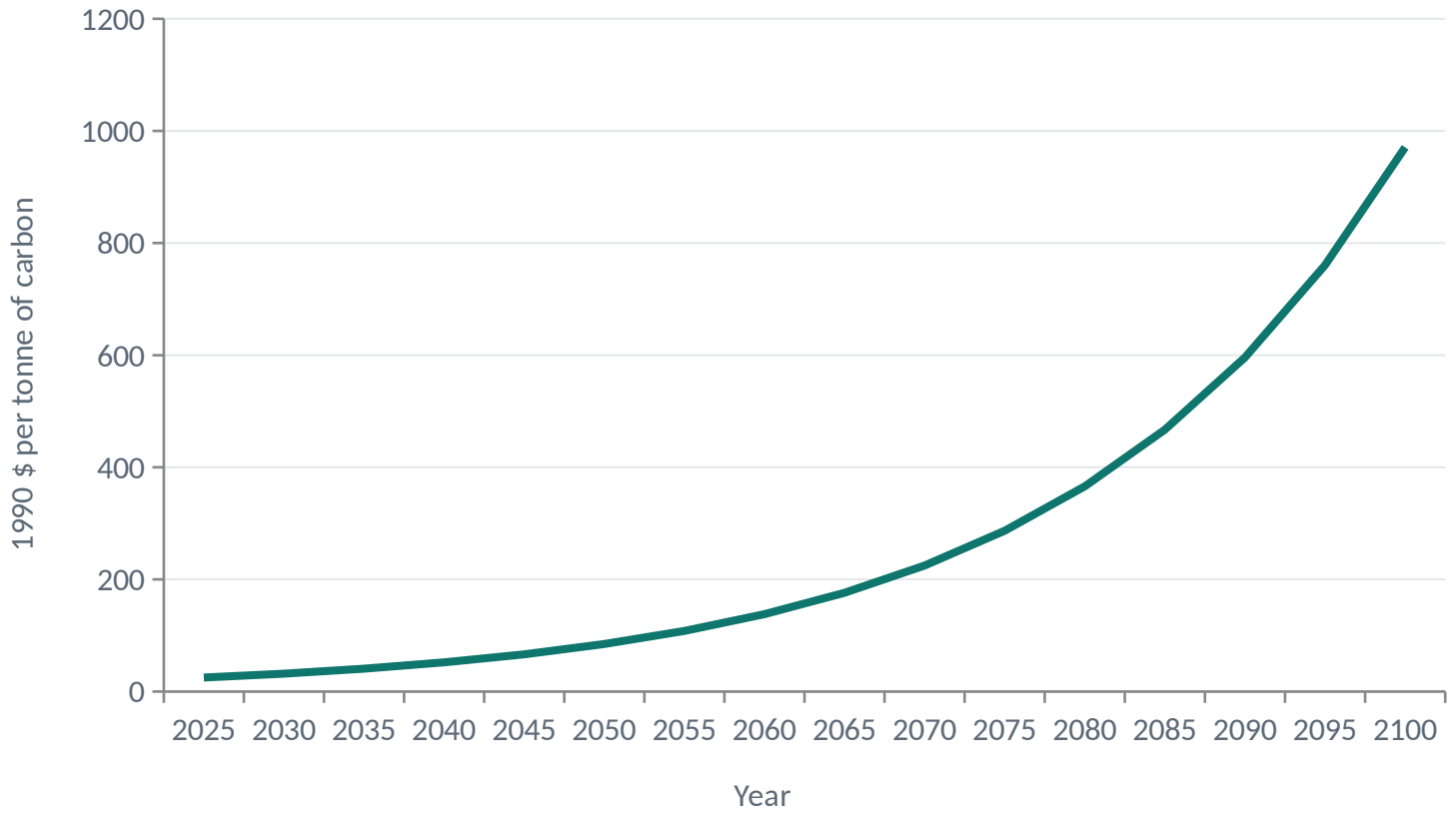
A toll road charges every car $5. You know the cost per car, but not how many will drive. A gate lets exactly 500 cars through. You know the number, but drivers bid up what a pass is worth. A carbon price is the toll; an emissions limit is the gate.

What it means for models

1. Price fixed → quantity (emissions) found
2. Cap fixed → price found
3. Same cost curve, read in opposite directions

PICTURE THE NUMBERS

The course's carbon price rises 5% a year



From carbon_price_global.xml used in lab 10, copied from GCAM's carbon_tax_25_5.xml

Read it like this

1. Starts at 25 in 2025, × 1.05 each year
2. 2050: 25 × 1.05^25 ≈ 85
3. Per tonne of CO2, divide by 3.67

Quick check

A carbon price starts at 25 and rises 5% a year. Roughly what is it 10 years later?

- **A** 30
- **B** 35
- **C** 41
- **D** 50

Quick check: answer

A carbon price starts at 25 and rises 5% a year. Roughly what is it 10 years later?

A 30

B 35

C 41 ✓

D 50

SIDE BY SIDE

Policy enters as a price, a limit or a rule

Policy type	You set	GCAM finds	Course scenario
Carbon price	A price each period	Emissions each period	CarbonPrice
Emissions limit	A cap each period	The price that meets it	NetZero
Climate target	A forcing level in a year	A least-cost price path	Not run: slow
Sector or land rule	A share, cap or protected land	The tax or subsidy needed	Not run

Regions that share a market name share one carbon price

Worked example: price or cap, the mirror

Problem Suppose each extra 1% cut costs $2 more per tonne, so price = 2 × cut%. A straight-line cost curve, for illustration.

1. CarbonPrice: you set $50
2. GCAM finds the cut: 50 ÷ 2 = 25% below Reference
3. NetZero-style cap: you require a 60% cut
4. GCAM finds the price: 2 × 60 = $120 per tonne

Answer

Price $50 → 25% cut; cap of 60% cut → price $120

KEY IDEA

Three ways GCAM removes CO2

1

Bioenergy with capture

Grow crops, burn them for energy, store the CO2. Needs land, water and fertiliser (week 9).

2

More forest

Trees store carbon as they grow. Needs land, and uptake takes 30 to 100 years.

3

Direct air capture

Machines pull CO2 from the air. Needs a lot of electricity or heat.

NUMBERS TO REMEMBER

Balancing a net-zero year

10 Gt

CO2 still emitted in 2050, to be balanced by removals

3 Gt

of that removed by direct air capture

18 EJ

energy needed: 3 × 10^9 t × 6 GJ per tonne

Illustrative split: 4 Gt bioenergy with capture + 3 Gt forest + 3 Gt air capture = 10 Gt

WATCH OUT

Common mix-ups

Not quite	Better
Net zero means no emissions	It means remaining emissions equal removals, and every removal uses land, energy or water.
GCAM has a built-in setting for each country's NDC	Turning a pledge into caps or prices is your modelling choice. State it in your report.
GCAM carbon prices are per tonne of CO2	They are in 1990 dollars per tonne of carbon. Divide by 3.67 for CO2.

NEW WORDS THIS WEEK

Words to add to the class glossary

Abatement Cutting emissions of a gas, for example capturing methane at a landfill. GCAM reads its cost from cost curves.

Carbon dioxide removal Taking CO2 out of the air and storing it, for example in forests, or with bioenergy plus capture, or machines (CDR).

Emissions limit A cap on how much may be emitted each period. The model finds the carbon price needed to meet it.

Net zero Remaining emissions equal removals, so the total added to the air is zero. It does not mean no emissions.

NDC Nationally Determined Contribution: a country's climate pledge under the Paris Agreement.

Lab 10: CarbonPrice, NetZero and gcamdata

Start a CarbonPrice run, compare CarbonPrice and NetZero with Reference for emissions, temperature and price, then change one number in gcamdata and rebuild.

Script: labs/lab10.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

Build the CarbonPrice run (Terminal tab)

Same routine as week 8: copy, make two edits, start. The add-on file is carbon_price_global.xml from your gcam-files folder.

```
cd ~/gcam/exe
cp configuration_ref.xml config_carbonprice.xml
# edit config_carbonprice.xml (two changes), then:
nohup ./gcam.exe -C config_carbonprice.xml &
tail -n 3 logs/main_log.txt
```



Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.

Add both policy scenarios (gap 1)

A for loop adds each scenario. Fill the gap with the second policy scenario's name.

```
prj <- loadProject("output/gcam_course.dat")
for (s in c("CarbonPrice", "___")) {  # 2nd policy
  prj <- addScenario(conn, prj, s,
    "gcam-files/course_queries.xml")
}
co2 <- getQuery(prj, "CO2 emissions by region") |>
  group_by(scenario, year) |>
  summarise(value = sum(value), .groups = "drop")
ggplot(co2, aes(year, value, colour = scenario)) +
  geom_line()
```

Temperature and carbon prices (gap 2)

Fill the gap with the query title that returns global mean temperature. Then look at the carbon price in each market.

```
temp <- getQuery(prj, "___")  # a query title
ggplot(temp, aes(year, value, colour = scenario)) +
  geom_line()
price <- getQuery(prj, "CO2 prices")
ggplot(price, aes(year, value, colour = scenario)) +
  geom_line() +
  facet_wrap(~ market)
```

⌨ *Fill each ___ gap before running that line. Stuck? The solution file has the complete version.*

Change one number in gcamdata (gap 3)

In your gcamdata project, raise solar's 2050 capacity factor in one CSV, rebuild only what depends on it, and compare.

```
devtools::load_all(".")
obj <- "L223.GlobalIntTechCapFac_elec"
before <- drake::readd(obj, character_only = TRUE)
file.edit(file.path("inst/extdata/energy",
  "A23.globaltech_capacity_factor.csv"))
# change the PV value for 2050, save, then:
devtools::load_all(".")
driver_drake()
after <- drake::readd(obj, character_only = TRUE)
filter(after, subsector.name == "solar",
       year == ___)  # the year you edited
```

LAB · CHECK YOUR WORK

Did it work?

You should see

- Three or four emission lines, with NetZero falling to about zero by 2050
- Temperature lines stay close until mid-century, then separate
- after shows your new 2050 capacity factor for solar

Worksheet questions

1. By how much are 2050 CO2 emissions below Reference in CarbonPrice and in NetZero?
2. Why do the temperature lines separate later than the emission lines? Use week 2's idea of cumulative emissions.
3. List the files driver_drake rebuilt after your edit. Which XML file would GCAM read?

RECAP

Take-home messages

1. GCAM counts emissions from activity; Hector turns them into warming
2. A carbon price fixes cost and finds emissions; a limit fixes emissions and finds the price
3. The course price starts at 25 (1990 $/tC) in 2025 and rises 5% a year
4. Net zero balances leftover emissions with removals, which use land, energy or water
5. Turning pledges into model inputs is a choice you must state

Next: week 11

Week 11 opens the advanced stage: optimisation models that look ahead, starting with a three-technology power system as a linear program.

Practise

Interactive course, week 10: try the carbonprice explorer and the 5-question quiz.

Readings: Fawcett et al. (2015), Can Paris pledges avert severe climate change?, Science; gcamdata guide: running the driver; GCAM v8.2 documentation: Emissions, Climate (Hector), Policies and costs, Policy examples